

MOHAMMADHOSSEIN ASKARIHEZAVEH

(Hossein Askari)

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Theartificialself.com
[GitHub](#) · [Portfolio](#)

PERSONAL PROFILE

Interdisciplinary researcher and creative technologist with a background in computational arts and AI. My work is focused on designing and analyzing multi-agent systems to explore the fundamental principles of emergent communication and language evolution. I am skilled in building experimental setups, analyzing complex data, and adapting quickly to new challenges.

RESEARCH INTERESTS

Language Evolution · Emergent Communication · Multi-Agent Systems · Computational Modeling · Artificial Intelligence · Social Semiotics

EDUCATION

Teacher Training Course (TTC) for English as a Second Language 2025
Safir Language Academy

MA in Computational Arts 2023–2024
Goldsmiths, University of London
Grade: Distinction
Thesis: "Exploring Socio-Cultural Behaviours and Language Evolution in AI"
Awards: Goldsmiths Computational Arts Scholarship for academic and creative excellence.

BSc in Architectural Engineering 2016–2021
Tehran Azad University
Final Grade: 15.57/20

RESEARCH AND WORK EXPERIENCE

English Teacher 2025–Present
Safir Language Academy

- Teaching English as a Second Language.

Creative Technologist & Research Collaborator 2024
Yaya Labs, London

- Designed and implemented a multi-agent communication system using AI avatars trained on complex, real-world linguistic data.
- Analyzed emergent narrative patterns and agent interaction in a live, dynamic environment.
- Investigated the effects of platform-level pressures (e.g., moderation) on agent communication strategies.

Master's Thesis Research 2023–2024
Goldsmiths, University of London

- Designed and implemented a multi-agent reinforcement learning system to study emergent communication.

- Developed AI-based evaluators to analyze agent-generated high-dimensional communication protocols.
- Applied high-dimensional language analysis to explore spontaneous narrative structuring under minimal constraints.

AI and Data Science Technician

2020–2022

Asan Darman Homecare

- Designed and deployed AI-driven predictive models for patient risk assessment, enabling dynamic resource allocation and triage optimization.
- Developed a machine learning pipeline integrating patient and staff data to forecast health-care demands, improving operational efficiency.

PUBLICATIONS

- **Knowledge Distillation Preserves Representational Geometry: Evidence from Coupled Transformer Dynamics and Attractor Analysis** (2026)
- **SM-MCPM: State-Modulated Monte Carlo Physarum Machine for Energy-Constrained Collective Navigation** (2025)
- **Cognitive Mesh: Tensor Product Fusion and Noise-Induced Emergence in Shared LLM Representations** (2025)
- **Coupled Neural Manifolds: Synchronization Dynamics in Transformer Embedding Spaces Without Linguistic Mediation: A Study of Pure Vector Dynamics in Large Language Model Architectures** (2024)
- **Emergence of High-Dimensional Communication Protocols in Multi-Agent Systems** (2024)

ACCEPTED WORKS

- **There's a party in my room and I'm not invited** | IEEE ICME 2025 | Exhibition and workshop talk

TECHNICAL SKILLS

- **Programming:** Python (TensorFlow, PyTorch, scikit-learn), C++ (OpenFrameworks), C# (Unity), JavaScript (Processing, P5.js).
- **Simulation & Modeling:** Multi-Agent Reinforcement Learning (MARL) System Design, Interactive 3D Environment Development (Unity, C#).
- **AI/ML:** Reinforcement Learning, Natural Language Processing, Computer Vision, Deep Learning.
- **Data Analysis:** Pandas, NumPy, Matplotlib, SQL.
- **Creative & Systems:** Web scraping, large-scale corpus building, generative AI, interactive systems, IoT, and sensor integration.

PROFESSIONAL MEMBERSHIPS

- Member, Multi-Agent Systems Group, The Alan Turing Institute.
- Member, Association d'Intelligence Artificielle France-Chine (AIFC).

REFEREES

- **Dr. Mattia Paganelli**, Lecturer, Goldsmiths University of London
m.paganelli@gold.ac.uk | [Academia Profile](#)
- **Jérémie Wenger**, Lecturer, Goldsmiths University of London
j.wenger@gold.ac.uk | [Personal Website](#)
- **Ben Ditto**, Creative Director, Yaya Labs
info@ditto-nation.com | [Yaya Labs Website](#)